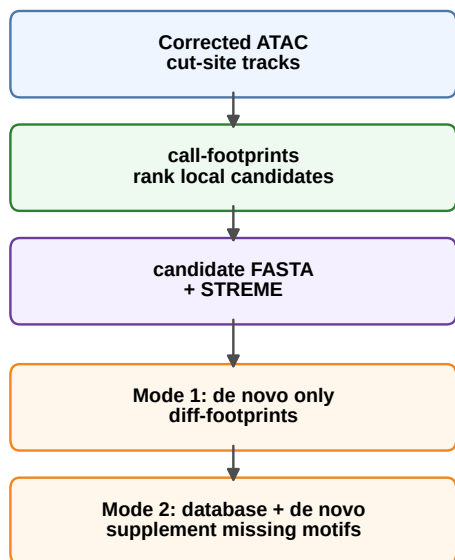


De novo motif discovery validates standalone and supplement modes

A. Two ways de novo motifs enter fp-tools

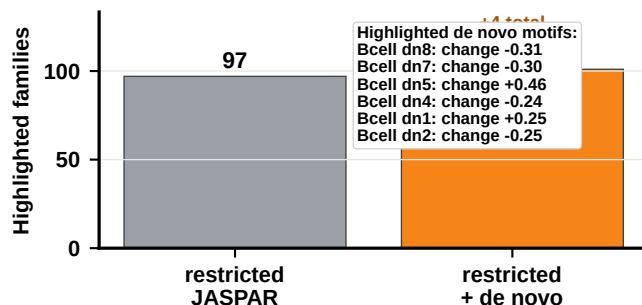


B. De novo-only discovery produces testable motif families

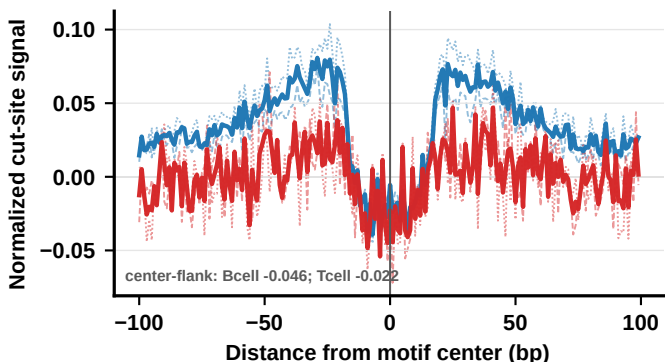
source	consensus	sites	Tomtom annotation
B candidates	GATGAGTCA	153	BATF (MA1634.2); q=2.8e-05
B candidates	AGGAACTGAAA	167	Irf1 (MA0050.4); q=7.6e-03
T candidates	AGGAAGTGACTG	28	IKZF2 (MA2326.1); q=1.1e-02
T candidates	CATTCTGGGAA	257	ZNF143 (MA0088.3); q=8.0e-05
B candidates	CCAGCCGGG	2,114	no confident match
B candidates	CAGGCCGCGCC	908	no confident match
B candidates	CCCGCGCC	789	no confident match

Tomtom labels are motif-similarity annotations, not definitive TF identity calls.

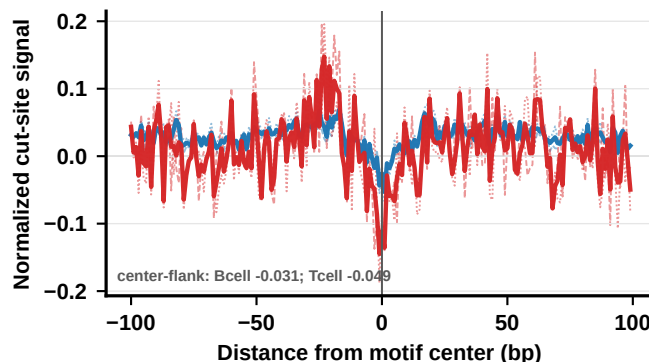
C. Supplement mode rescues extra differential motif families



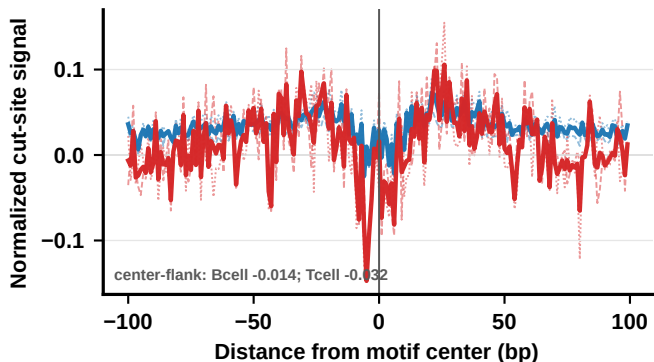
D. Bcell_denovo_5 (Bcell_denovo_5_5-GATGAGTCA)
B-cell-deeper de novo footprint; n=9,974; Bcell deeper



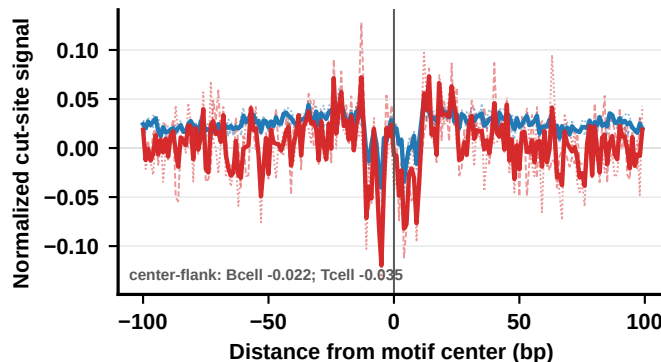
E. Tcell_denovo_4 (Tcell_denovo_4_4-GAHGYGGAA)
T-cell-deeper de novo footprint; n=4,858; Tcell deeper



F. Tcell_denovo_6 (Tcell_denovo_6_6-AGGAAGTSACTGA)
T-cell-deeper de novo footprint; n=9,294; Tcell deeper



G. Tcell_denovo_1 (Tcell_denovo_1_1-ACAGTTTCCT)
weaker T-cell-deeper footprint; n=9,820; Tcell deeper



..... Bcell_rep1 - - - Bcell_rep2 — Bcell mean Tcell_rep1 - - - Tcell_rep2 — Tcell mean